

Homework 3

MATH 518 - Fall 2025

Due on Sunday, November 2, 11:59pm on Crowdmark

Total of points = 40 points

Please submit each question separately on Crowdmark. Unless stated otherwise, k is an algebraically closed field of characteristic zero.

Recall that we can embed the affine space in the projective space by the maps

$$\varphi_j: k^n \longrightarrow U_j := \{[x_1 : \cdots : x_{n+1}] \mid x_j \neq 0\} \subset \mathbb{P}^n(k),$$

for $1 \leq j \leq n+1$. The complement $H_\infty = \mathbb{P}^n(k) \setminus U_{n+1}$ is called the hyperplane at infinity. Given an affine algebraic set $V \subset k^n$, we defined the projective closure $V^* \subset \mathbb{P}^n(k)$ by $V^* = V(I(V)^*)$. It is a projective algebraic set that satisfies $\varphi_{n+1}(V) = V^* \cap U_{n+1}$. Conversely, given $W \subset \mathbb{P}^n(k)$, we define the affine algebraic set $W_* = V(I(W)_*) = W \cap U_{n+1}$. We have

- $(V^*)_* = V$ for any affine algebraic set $V \subset k^n$,
- $(W_*)^* = W$ for any projective algebraic set $W \subset \mathbb{P}^n(k)$ such that W is not contained in H_∞ nor contains H_∞ .

From this and the Exercise 3 below, it follows that the maps $V \longmapsto V^*$ and $W \longmapsto W_*$ induce a one-to-one correspondence between projective varieties $W \subset \mathbb{P}^n(k)$ that are not contained in H_∞ , and the affine varieties in k^n . You can use this in Exercise 4.

1. (10 points) Let $H_1, \dots, H_m$ be hyperplanes in $\mathbb{P}^n(k)$. If $m \leq n$, show that

$$H_1 \cap \cdots \cap H_m \neq \emptyset.$$

Solution: Let $H_i = V_p(L_i)$ where $L_i \in k[x_1, \dots, x_{n+1}]$ is a homogeneous polynomial of degree 1. Since H_i is non-empty, the affine cone over H_i is $C(H_i) = V_a(L_i) \subset k^{n+1}$. It is a hyperplane in k^{n+1} passing through the origin. The intersection of these m hyperplanes is the solution in k^{n+1} of the system of m linear equations

$$\begin{aligned} L_1(x_1, \dots, x_{n+1}) &= 0 \\ L_2(x_1, \dots, x_{n+1}) &= 0 \\ &\vdots \\ L_m(x_1, \dots, x_{n+1}) &= 0. \end{aligned}$$

Let r be the rank of the system. The dimension of the space of solutions is $n + 1 - r \geq n + 1 - m \geq 1$, since we assumed that $m \leq n$. Thus, the intersection $C(H_1) \cap \cdots \cap C(H_m)$ contains at least a line l passing through the origin, and the corresponding point $p \in \mathbb{P}^n(k)$ lies in $H_1 \cap \cdots \cap H_m$. In particular, it is non-empty.

2. (10 points) Let $V = V(f_1, f_2) \subset k^3$, where $f_1, f_2 \in k[x, y, z]$ are the polynomials given by $f_1 = y - x^2$ and $f_2 = z - x^3$. Let $I(V)^* \subset k[x, y, z, w]$ be the homogenization of the ideal $I(V) = (f_1, f_2)$. Show that $zw - xy \in I(V)^*$ but $zw - xy \notin (f_1^*, f_2^*)$.

We always have $(f_1^, \dots, f_r^*) \subset (f_1, \dots, f_r)^*$, but this example shows that we don't always have equality. However, for principal ideals we have seen in class that $(f)^* = (f^*)$.*

Solution: The polynomial $g := f_2 - xf_1 = z - xy$ is in $I(V)$, thus $g^* = zw - xy \in I(V)^*$ is in the homogenization of $I(V)$.

Let us show that $zw - xy \notin (f_1^*, f_2^*)$. Note that $f_1^* = wy - x^2$ and $f_2^* = zw^2 - x^3$. Suppose that we had $a, b \in k[x, y, z, w]$ such that

$$zw - xy = af_1^* + bf_2^* = a(wy - x^2) + b(zw^2 - x^3).$$

By writing $a = a_0 + a_1 + \cdots + a_n$ and b into homogeneous degrees, we see that the homogeneous piece of degree 2 in the RHS is $a_0(wy - x^2)$, which is not equal to $zw - xy$.

3. (10 points) Let $V \subset W \subset \mathbb{P}^n(k)$ be two (non-empty) projective varieties, where $V = V(f)$ is a hypersurface defined by a homogeneous irreducible polynomial f . Show that $W = V$ or $W = \mathbb{P}^n(k)$. Deduce that the projective varieties $W \subset \mathbb{P}^n(k)$ that contain H_∞ are $W = \mathbb{P}^n(k)$ and $W = H_\infty$. Determine W_* and $(W_*)^*$ in the two cases.

Solution: Suppose that $V(f) \subset W \subset \mathbb{P}^n(k)$. Since these are non-empty, by the projective Nullstellensatz we have

$$(0) \subset I(W) \subset (f),$$

where $I(W)$ is prime. If $(0) = I(W)$, then $W = \mathbb{P}^n(k)$. On the other hand, if $f \in I(W)$, this would show that $I(W) = (f)$.

So let's assume that there exists a projective variety W for which $(0) \neq I(W)$ and $f \notin I(W)$. Let $g \in I(W)$ be a nonzero polynomial of minimal degree (which exists by our first assumption). Since $I(W) \subset (f)$, we can write $g = fh$ for some $h \in k[x_1, \dots, x_{n+1}]$. The ideal $I(W)$ is prime and we assumed that $f \notin I(W)$, from which we deduce that $h \in I(W)$. Moreover, we have $\deg(h) < \deg(g)$, since $\deg(f) > 0$. This contradicts the minimality of the degree of g , hence such a projective variety W does not exist.

In particular, this applies to the hypersurface $H_\infty = V(x_{n+1})$, and we deduce that if $W \subset \mathbb{P}^n(k)$ contains H_∞ , then $W = \mathbb{P}^n(k)$ or $W = H_\infty$. Finally, in these two cases we have $W_* = k^{n+1}$ if $W = \mathbb{P}^n(k)$, and $W_* = \emptyset$ if $W = H_\infty$. So $(W_*)^* = \mathbb{P}^n(k) = W$ if $W = \mathbb{P}^n(k)$, and $(W_*)^* = \emptyset$ if $W = H_\infty$.

4. (10 points) Describe all projective varieties in $\mathbb{P}^1(k)$ and $\mathbb{P}^2(k)$.

Solution: When $\mathbb{P}^1(k)$, the hyperplane at infinity is just the point $H_\infty = [1 : 0]$. By the previous correspondence, the other projective varieties correspond to affine varieties in k , which are just points, or k itself. So the projective varieties in $\mathbb{P}^1(k)$ are $\mathbb{P}^1(k)$ itself, and points.

For $\mathbb{P}^2(k)$, the hyperplane at infinity is $H_\infty = \{[x_1 : x_2 : 0]\} \simeq \mathbb{P}^1(k)$. The varieties contained in H_∞ have been described by the first part. They are points, or H_∞ itself (which is the curve $V(x_3)$). For the other projective varieties not contained in H_∞ , they correspond to affine varieties in k^2 . We have seen that these can be points, irreducible curves $V(f)$, and k^2 itself. Since $V(f)^* = V(f^*)$, the projective varieties in $\mathbb{P}^2(k)$ are points, irreducible curves $V(F)$ with F homogeneous, and $\mathbb{P}^2(k)$ itself.